



BACHELOR IN INFORMATION TECHNOLOGY (HONOURS)

FINAL EXAMINATION FEBRUARY 2025

Course: EC3103 (Introduction to Information Systems)

Time: 8.00 am – 11.00 am (3 hours)

Lecturer: Muhammad Hamizan Johari

Date: 23 May 2025

Instructions:

Answer **ALL** questions.

The maximum number of marks is 100.

This question paper consists of 3 printed pages.
(excluding front cover)

DO NOT OPEN THIS BOOKLET UNTIL YOU ARE TOLD TO DO SO

1. a. Briefly discuss **FOUR (4)** foundation concepts of information systems. (8 MARKS)
 - b. Briefly describe **TWO (2)** ways to process transactions. (4 MARKS)
 - c. Differentiate the **THREE (3)** types of operations support systems based on their main functions. (9 MARKS)
 - d. Distinguish the appropriate management support systems based on the given functions.
 - i. Critical information for executives and managers. (2 MARKS)
 - ii. Interactive and ad-hoc support. (2 MARKS)
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2. a. Briefly explain **THREE (3)** strategies a company can apply to achieve agility. (6 MARKS)
 - b. Briefly explain **THREE (3)** basic virtual strategies a company can apply to help their business. (6 MARKS)

The business use of information technology refers to the strategic application of IT tools and systems to enhance various aspects of business operations and decision-making. It enables companies to lower operational costs, differentiate their products and services, innovate through new offerings and improved processes, promote business growth across regions and markets, and develop strong alliances with partners and stakeholders. By leveraging technologies such as data analytics, enterprise systems, cloud computing, and communication networks, businesses can improve efficiency, gain competitive advantages, respond quickly to market changes, and deliver greater value to customers and partners.

- c. Based from the given paragraph, analyze and suggest at least **THREE (3)** ways to implement basic strategies in the business use of information technology. (9 MARKS)
- d. Categorize the knowledge involved in the given scenario below.
 - i. At a logistics company, a new employee is assigned to manage shipping procedures. To get started, they are given access to a shared drive containing detailed documentation, process flowcharts, Excel tracking sheets, and company policies. Everything they need to perform their tasks is written down and stored in the system, allowing them to follow step-by-step instructions with minimal supervision. (2 MARKS)

online
Transaction

- ii. In a manufacturing plant, a senior technician has mastered how to fine-tune a specific machine to reduce defects — something not found in the manuals. This "know-how" is based on years of hands-on experience and intuition. When mentoring a junior staff member, the technician explains the subtle sounds and vibrations to listen for and demonstrates adjustments in real time. **(2 MARKS)**
3. a. Briefly discuss ethics in information systems. **(6 MARKS)**
- b. Briefly describe the following ethical theories related to moral principles in information systems.
- i. Deontological Ethics (Duty-Based) **(3 MARKS)**
 - ii. Utilitarian Ethics (Outcome-Based): **(3 MARKS)**

In a remote village in Southeast Asia, access to quality education is limited due to a lack of infrastructure and trained teachers. To bridge this gap, the government partners with the One Laptop per Child (OLPC) initiative to distribute low-cost, durable laptops to every primary school student. These laptops come preloaded with educational software, interactive lessons, and basic coding programs that can run without internet. For the first time, children can explore digital content, practice math and language skills, and even collaborate on projects using wireless mesh networking. Over time, the laptops empower the students to become more curious, tech-savvy, and better prepared for further education, breaking the cycle of digital and educational inequality.

- c. Analyze the paragraph above and state **TWO (2)** possible shortcomings of the initiative. **(4 MARKS)**
- d. Differentiate data, information and knowledge based from their definition. **(9 MARKS)**
4. a. Briefly explain the following emerging threats caused by advancements in information systems technologies.
- i. Internet of Things (IoT) **(2 MARKS)**
 - ii. Artificial Intelligence (AI) **(2 MARKS)**
- b. Briefly discuss **THREE (3)** main importance of security in business. **(6 MARKS)**
- c. Briefly describe system vulnerability. **(2 MARKS)**

Investing in security delivers significant business value by safeguarding sensitive data, protecting against cyberattacks, and minimizing the risk of financial loss, reputational damage, and legal liabilities. It helps ensure business continuity by preventing disruptions caused by security breaches and supports regulatory compliance with industry standards and data protection laws. Furthermore, strong security measures build customer trust and confidence, which are essential for maintaining long-term relationships and competitive advantage in a digitally connected marketplace.

- d. Analyze the given paragraph and state the importance of investing in security to increase business value. (9 MARKS)
- e. Differentiate at least **TWO (2)** goals of information security. (4 MARKS)

-END OF QUESTION PAPER-